

Exercise: 14.3

Partial Derivatives:

$$f(u, y) = 2u^2 - 3y - 4$$

w.r.t u

$$\frac{\partial f}{\partial u} = 2 \frac{\partial}{\partial u} (u^2) - 3y \frac{\partial}{\partial u} (1) - 4 \frac{\partial}{\partial u} (1)$$

$$= 2(2u) - 0 - 0$$

$$\frac{\partial f}{\partial u} = 4u$$

w.r.t y

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (2u^2) - 3 \frac{\partial}{\partial y} (y) - \frac{\partial}{\partial y} (4)$$

$$= 0 - 3 - 0$$

$$\frac{\partial f}{\partial y} = -3$$

$$f(u, y) = u^2 - uy + y^2$$

w.r.t u

$$\frac{\partial f}{\partial u} = \frac{\partial}{\partial u} (u^2) - y \frac{\partial}{\partial u} (u) + \frac{\partial}{\partial u} (y^2)$$

$$= 2u - y + 0$$

$$\frac{\partial f}{\partial u} = 2u - y$$

w.r.t y

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (u^2) - u \frac{\partial}{\partial y} (y) + \frac{\partial}{\partial y} (y^2)$$
$$= 0 - u + 2y$$

$$\frac{\partial f}{\partial y} = 2y - u$$

$$f(u, y) = (u^2 - 1)(y + 2)$$

w.r.t u

$$f(u, y) = u^2 y + 2u^2 - y - 2$$

$$\frac{\partial f}{\partial u} = y \frac{\partial}{\partial u} (u^2) + 2 \frac{\partial}{\partial u} (u^2) - \frac{\partial}{\partial u} (y) - \frac{\partial}{\partial u} (2)$$
$$= 2uy + 4u - 0 - 0$$

$$\frac{\partial f}{\partial u} = 2u(y + 2)$$

w.r.t y

$$\frac{\partial f}{\partial y} = u^2 \frac{\partial}{\partial y} (y) + \frac{\partial}{\partial y} (2u^2) - \frac{\partial}{\partial y} (y) - \frac{\partial}{\partial y} (2)$$
$$= u^2 + 0 - 1 - 0$$

$$\frac{\partial f}{\partial y} = u^2 - 1$$

$$f(u, y) = 5uy - 7u^2 + y^2 + 3u - 6y + 2$$

w.r.t u

$$\frac{\partial f}{\partial u} = 5y - 14u + 0 + 3 - 0 + 0$$

∂u

$$\frac{\partial f}{\partial u} = 5y - 14u + 3$$

∂u

w.r.t y

$$\frac{\partial f}{\partial y} = 5u - 0 + 2y + 0 - 6 + 0$$

∂y

$$\frac{\partial f}{\partial y} = 5u + 2y - 6$$

∂y

$$f(u, y) = (uy - 1)^2$$

$$f(u, y) = u^2y^2 - 2uy + 1$$

w.r.t u

$$\frac{\partial f}{\partial u} = 2uy^2 - 2y$$

∂u

$$\frac{\partial f}{\partial u} = 2y(uy - 1)$$

∂u

w.r.t y

$$\frac{\partial f}{\partial y} = 2u^2y - 2u$$

∂y

$$\frac{\partial f}{\partial y} = 2u(uy - 1)$$

∂y

$$f(x, y) = (2x - 3y)^3$$

w.r.t x

$$\frac{\partial f}{\partial x} = 3(2x - 3y)^2 (2)$$

$\frac{\partial f}{\partial x}$

$$\frac{\partial f}{\partial x} = 6(2x - 3y)^2$$

$\frac{\partial f}{\partial x}$

w.r.t y

$$\frac{\partial f}{\partial y} = 3(2x - 3y)^2 (-3)$$

$\frac{\partial f}{\partial y}$

$$\frac{\partial f}{\partial y} = -9(2x - 3y)^2$$

$\frac{\partial f}{\partial y}$

$$f(x, y) = \sqrt{x^2 + y^2}$$

w.r.t x

$$\frac{\partial f}{\partial x} = \frac{1}{\sqrt{x^2 + y^2}} (2x)$$

$$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}$$

w.r.t y

$$\frac{\partial f}{\partial y} = \frac{1}{\sqrt{x^2 + y^2}} (2y)$$

$$\frac{\partial f}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial f}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial f}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$f(u, y) = \left(u^3 + \frac{y}{2} \right)^{2/3}$$

w.r.t u

$$\frac{\partial f}{\partial u} = \frac{2}{3} \left(u^3 + \frac{y}{2} \right)^{-1/3} (3u^2)$$

$$\frac{\partial f}{\partial u} = 2u^2 \left(u^3 + \frac{y}{2} \right)^{-1/3}$$

w.r.t y

$$\frac{\partial f}{\partial y} = \frac{2}{3} \left(u^3 + \frac{y}{2} \right)^{-1/3} \left(\frac{1}{2} \right)$$

$$\frac{\partial f}{\partial y} = \frac{1}{3} \left(u^3 + \frac{y}{2} \right)^{-1/3}$$

$$f(u, y) = \frac{1}{u+y}$$

u+y

w.r.t u

$$\frac{\partial f}{\partial u} = (-1)(u+y)^{-2}$$

u

$$\frac{\partial f}{\partial u} = \frac{-1}{(u+y)^2}$$

$$\frac{\partial f}{\partial u} = \frac{-1}{(u+y)^2}$$

w.r.t y

$$\frac{\partial f}{\partial y} = (-1)(u+y)^{-2}$$

y

$$\frac{\partial f}{\partial y} = \frac{-1}{(u+y)^2}$$

$$\frac{\partial f}{\partial y} = \frac{-1}{(u+y)^2}$$

$$f(u, y) = \frac{u}{u^2 + y^2}$$

w.r.t u

$$\begin{aligned}\frac{\partial f}{\partial u} &= \frac{(u^2 + y^2)(1) - u(2u)}{(u^2 + y^2)^2} \\ &= \frac{u^2 + y^2 - 2u^2}{(u^2 + y^2)^2}\end{aligned}$$

$$\frac{\partial f}{\partial u} = \frac{y^2 - u^2}{(u^2 + y^2)^2}$$

w.r.t y

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{(u^2 + y^2)(0) - u(2y)}{(u^2 + y^2)^2} \\ \frac{\partial f}{\partial y} &= \frac{-2uy}{(u^2 + y^2)^2}\end{aligned}$$

$$f(u, y) = \frac{(u+y)}{(u-y-1)}$$

w.r.t u

$$\begin{aligned}\frac{\partial f}{\partial u} &= \frac{(u-y-1)(1) - (u+y)(-1)}{(u-y-1)^2} \\ &= \frac{(u-y-1) + (u+y)}{(u-y-1)^2} \\ &= \frac{\cancel{u}y-1-\cancel{u}y-y^2}{(u-y-1)^2}\end{aligned}$$

$$\frac{\partial f}{\partial u} = \frac{-y^2 - 1}{(uy-1)^2}$$

w.r.t y

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{(uy-1)(1) - (u+y)(u)}{(uy-1)^2} \\ &= \frac{uy-1-u^2-uy}{(uy-1)^2}\end{aligned}$$

$$\frac{\partial f}{\partial y} = \frac{-1-u^2}{(uy-1)^2}$$

$$f(u, y) = \tan^{-1}\left(\frac{u}{y}\right)$$

w.r.t u

$$\begin{aligned}\frac{\partial f}{\partial u} &= \frac{1}{1 + (y/u)^2} \cdot \left(\frac{-y}{u^2}\right) \\ &= \frac{\cancel{u}}{u^2 + y^2} \left(\frac{-y}{\cancel{u}}\right)\end{aligned}$$

$$\frac{\partial f}{\partial u} = \frac{-y}{u^2 + y^2}$$

w.r.t y

$$\frac{\partial f}{\partial y} = \frac{\cancel{u}}{u^2 + y^2} \left(\frac{1}{\cancel{y}}\right)$$

$$\frac{\partial f}{\partial y} = \frac{u}{u^2 + y^2}$$

$$f(u, y) = e^{(u+y+1)}$$

w.r.t u

$$\frac{\partial f}{\partial u} = e^{(u+y+1)} \quad (1)$$

$\frac{\partial f}{\partial u}$

$$\frac{\partial f}{\partial u} = e^{(u+y+1)}$$

$\frac{\partial f}{\partial u}$

w.r.t y

$$\frac{\partial f}{\partial y} = e^{(u+y+1)} \quad (1)$$

$\frac{\partial f}{\partial y}$

$$\frac{\partial f}{\partial y} = e^{u+y+1}$$

$\frac{\partial f}{\partial y}$

$$f(u, y) = e^{-u} \sin(u+y)$$

w.r.t u

$$\frac{\partial f}{\partial u} = e^{-u}(-1) \sin(u+y) + \cos(u+y) e^{-u}$$

$$\frac{\partial f}{\partial u} = +e^{-u}(-\sin(u+y) + \cos(u+y))$$

w.r.t y

$$\frac{\partial f}{\partial y} = e^{-u} \cos(u+y)$$

$$f(u, y) = \ln(u+y)$$

w.r.t u

$$\frac{\partial f}{\partial u} = \frac{1}{u+y}$$

w.r.t y

$$\frac{\partial f}{\partial y} = \frac{1}{u+y}$$

$$f(u, y) = e^{uy} \ln y$$

w.r.t u

$$\frac{\partial f}{\partial u} = \ln y \cdot e^{uy} \cdot y$$

$$= y \cdot e^{uy} \ln y$$

w.r.t y

$$\frac{\partial f}{\partial y} = e^{uy} \cdot 1 + \ln y \cdot e^{uy} \cdot u$$

$$\frac{\partial f}{\partial y} = \frac{1}{y} e^{uy} + u e^{uy} \ln y$$

$$f(u, y) = \sin^2(u-3y)$$

$$\frac{\partial f}{\partial u} = 2 \sin(u-3y) \cos(u-3y)$$

$$\frac{\partial f}{\partial y} = 2 \sin(u-3y) \cos(u-3y) (-3)$$

$$\frac{\partial f}{\partial y} = -6 \sin(u-3y) \cos(u-3y)$$

$$f(u, y) = \cos^2(3u - y^2)$$

$$\frac{\partial f}{\partial u} = 2 \cos(3u - y^2) (-\sin(3u - y^2)) 3$$

$$\frac{\partial f}{\partial u} = -6 \cos(3u - y^2) \sin(3u - y^2)$$

$$\frac{\partial f}{\partial y} = 2 \cos(3u - y^2) (-\sin(3u - y^2)) (-2y)$$

$$= 4y \cos(3u - y^2) \sin(3u - y^2)$$

$$f(u, y) = u^y$$

$$\frac{\partial f}{\partial u} = y u^{y-1}$$

$$\frac{\partial f}{\partial y} = u^y \ln u$$

$$f(u, y) = \log y^u$$

$$\frac{\partial f}{\partial u} = \frac{1}{y \ln y}$$

$$\frac{\partial f}{\partial y} = \ln u \left(\frac{1}{y \ln y} \right)^{-1}$$

$$\frac{\partial f}{\partial y} = -\frac{\ln u}{y (\ln y)^2}$$

Question 23-34

$$f(x, y, z) = 1 + xy^2 - 2z^2$$

w.r.t x

$$f_x = 0 + y^2 - 0 = y^2$$

w.r.t y

$$f_y = 0 + 2xy - 0 = 2xy$$

w.r.t z

$$f_z = 0 + 0 - 4z$$

$$= -4z$$

$$f(x, y, z) = xyz + yz + xz$$

w.r.t x

$$f_x = y + 0 + z = y + z$$

w.r.t y

$$f_y = x + z$$

w.r.t z

$$f_z = y + x$$

$$f(x, y, z) = x - \sqrt{y^2 + z^2}$$

w.r.t x

$$f_x = 1$$

w.r.t y

$$f_y = 0 - \frac{1}{2} \frac{(2y)}{\sqrt{y^2 + z^2}} = \frac{-y}{\sqrt{y^2 + z^2}}$$

w.r.t. z

$$f_z = \frac{-1}{2\sqrt{y^2+z^2}} (2z) = \frac{-z}{\sqrt{y^2+z^2}}$$

$$f(u, y, z) = (u^2 + y^2 + z^2)^{-1/2}$$

w.r.t. u

$$f_u = \frac{-1}{2} (u^2 + y^2 + z^2)^{-3/2} (2u)$$

$$f_u = \frac{-u}{(u^2 + y^2 + z^2)^{3/2}}$$

w.r.t. y

$$f_y = \frac{-1}{2} (u^2 + y^2 + z^2)^{-3/2} (2y)$$

$$f_y = \frac{-y}{(u^2 + y^2 + z^2)^{3/2}}$$

w.r.t. z

$$f_z = \frac{-1}{2} (u^2 + y^2 + z^2)^{-3/2} (2z)$$

$$f_z = \frac{-z}{(u^2 + y^2 + z^2)^{3/2}}$$

$$f(u, y, z) = \sin^{-1}(uyz)$$

w.r.t. u

$$f_u = \frac{1}{\sqrt{1-(uyz)^2}} (yz)$$

w.r.t. y

$$f_y = \frac{1}{\sqrt{1-(uyz)^2}} (uz)$$

w.r.t. z

$$f_z = \frac{1}{\sqrt{1-(uyz)^2}} (uy)$$

$$f(u, y, z) = \sec^{-1}(u+yz)$$

w.r.t. u

$$f_u = \frac{1}{|u+yz| \sqrt{(u+yz)^2 - 1}}$$

w.r.t. y

$$f_y = \frac{1}{|u+yz| \sqrt{(u+yz)^2 - 1}} (z)$$

w.r.t. z

$$f_z = \frac{1}{|u+yz| \sqrt{(u+yz)^2 - 1}} (y)$$

$$f(u, y, z) = \ln(u + 2y + 3z)$$

w.r.t u

$$f_u = \frac{1}{u + 2y + 3z}$$

$$u + 2y + 3z$$

w.r.t y

$$f_y = \frac{1}{u + 2y + 3z} \quad (2)$$

$$u + 2y + 3z$$

w.r.t z

$$f_z = \frac{1}{u + 2y + 3z} \quad (3)$$

$$u + 2y + 3z$$

$$f(u, y, z) = yz \ln(uy)$$

w.r.t u

$$f_u = \frac{yz}{uy} = \frac{yz}{u}$$

$$uy \quad u$$

w.r.t y

$$f_y = \frac{yz}{uy} + \ln(uy) \quad (z)$$

$$(uy)$$

$$= \frac{z}{u} + z \ln(uy)$$

w.r.t z

$$f_z = \ln(uy) \quad (y)$$

$$= y \ln(uy)$$

$$f(u, y, z) = e^{-(u^2 + y^2 + z^2)}$$

w.r.t u

$$\begin{aligned} f_u &= e^{-(u^2 + y^2 + z^2)} (-2u) \\ &= -2ue^{-(u^2 + y^2 + z^2)} \end{aligned}$$

w.r.t y

$$f_y = -2y e^{-(u^2 + y^2 + z^2)}$$

w.r.t z

$$f_z = -2z e^{-(u^2 + y^2 + z^2)}$$

$$f(u, y, z) = e^{-xyz}$$

w.r.t u

$$\begin{aligned} f_u &= e^{-xyz} (-yz) \\ &= -yz e^{-xyz} \end{aligned}$$

w.r.t y

$$f_y = -xz e^{-xyz}$$

w.r.t z

$$f_z = -xy e^{-xyz}$$

$$f(u, y, z) = \tanh(u + 2y + 3z)$$

w.r.t u

$$f_u = \operatorname{Sec}^2(u + 2y + 3z)$$

w.r.t y

$$f_y = \operatorname{Sec}^2(u + 2y + 3z)(2)$$

w.r.t z

$$f_z = \operatorname{Sec}^2(u + 2y + 3z)(3)$$

$$f(u, y, z) = \sinh(uy - z^2)$$

w.r.t u

$$f_u = \cosh(uy - z^2)(y)$$

w.r.t y

$$f_y = \cosh(uy - z^2)(u)$$

w.r.t z

$$f_z = \cosh(uy - z^2)(-2z) \\ = -2z \cosh(uy - z^2)$$

Find 2nd order derivative.

41-46

$$f(u, y) = u + y + uy$$

w.r.t u

$$f_u = 1 + y$$

$$f_{uu} = 0 + 0 = 0$$

$$f_{uy} = 0 + 1 = 1$$

w.r.t y

$$f_y = 1 + u$$

$$f_{yy} = 0 + 0 = 0$$

$$f_{yu} = 0 + 1 = 1$$

$$f(u, y) = \sin uy$$

w.r.t u

$$f_u = y \cos uy$$

$$f_{uu} = -y^2 \sin uy$$

$$f_{uy} = -y \sin(uy)(u) + \cos uy \\ = -uy \sin(uy) + \cos uy$$

w.r.t y

$$f_y = u \cos y$$

$$f_{yy} = -u^2 \sin y$$

$$\begin{aligned} f_{yu} &= -u \sin y (y) + \cos y \\ &= -u y \sin y + \cos y \end{aligned}$$

$$g(u, y) = u^2 y + \cos y + y \sin u$$

w.r.t u

$$g_u = 2uy + 0 + y \cos u$$

$$g_{uu} = 2y - y \sin u$$

$$g_{uy} = 2u + \cos u$$

w.r.t y

$$g_y = u^2 - \sin y + \sin u$$

$$g_{yy} = 0 - \cos y + \cos u (0)$$

$$g_{yu} = 2u + \cos y$$

$$h(u, y) = ue^y + y + 1$$

w.r.t u

$$h_u = e^y + 0 + 0$$

$$h_{uu} = 0$$

$$h_{uy} = e^y$$

w.r.t y

$$h_y = ue^y + 1$$

$$h_{yy} = ue^y$$

$$\begin{aligned} h_{yu} &= 0 + e^y \\ &= e^y \end{aligned}$$

$$r(u, y) = \ln(u+y)$$

w.r.t u

w.r.t y

$$r_u = \frac{1}{u+y}$$

$$r_y = \frac{1}{u+y}$$

$$r_{uu} = \frac{-1}{(u+y)^2}$$

$$r_{yy} = \frac{-1}{(u+y)^2}$$

$$r_{uy} = \frac{-1}{(u+y)^2}$$

$$r_{yu} = \frac{-1}{(u+y)^2}$$

$$r_{uy} = \frac{-1}{(u+y)^2}$$

$$r_{yu} = \frac{-1}{(u+y)^2}$$

$$r_{uy} = \frac{-1}{(u+y)^2}$$

$$r_{yu} = \frac{-1}{(u+y)^2}$$

$$S(u, y) = \tan^{-1}\left(\frac{y}{u}\right)$$

w.r.t u

$$S_u = \frac{1}{1 + \frac{y^2}{u^2}} \left(\frac{-y}{u^2} \right)$$

$$S_u = \frac{-y}{u^2 + y^2}$$

$$S_{uu} = \frac{(u^2 + y^2)(-y)' - (-y)(u^2 + y^2)'}{(u^2 + y^2)^2}$$

$$S_{uu} = \frac{2uy}{(u^2 + y^2)^2}$$

$$S_{uy} = \frac{(u^2 + y^2)(-y)' - (-y)(u^2 + y^2)'}{(u^2 + y^2)^2}$$

$$= \frac{(u^2 + y^2)(-1) - (-y)(2y)}{(u^2 + y^2)^2}$$

$$= \frac{-u^2 - y^2 + 2y^2}{(u^2 + y^2)^2}$$

$$S_{xy} = \frac{y^2 - u^2}{(u^2 + y^2)^2}$$

w.r.t y

$$S_y = \frac{1}{1 + y^2/u^2} \left(\frac{1}{u} \right)$$

$$= \frac{u}{u^2 + y^2}$$

$$S_{yy} = \frac{(u^2 + y^2)(u)' - (u)(u^2 + y^2)'}{(u^2 + y^2)^2}$$

$$= \frac{(u^2 + y^2)(0) - 2uy}{(u^2 + y^2)^2} \Rightarrow \frac{-2uy}{(u^2 + y^2)^2}$$

$$S_{yu} = \frac{(u^2 + y^2)(u)' - (u)(u^2 + y^2)'}{(u^2 + y^2)^2}$$

$$= \frac{\cancel{2uy} - 2u^2}{(u^2 + y^2)^2}$$

$$S_{yu} = \frac{y^2 - u^2}{(u^2 + y^2)^2}$$

47-50.

verify $w_{xy} = w_{yx}$

$$w = \ln(2x + 3y)$$

w.r.t x

$$w_x = \frac{1}{2x + 3y} \quad (2)$$

$$2x + 3y$$

w.r.t y

$$w_y = \frac{1}{2x + 3y} \quad (3)$$

$$2x + 3y$$

$$w_{xy} = \frac{-2}{(2x+3y)^2} \quad (3)$$

$$w_{yx} = \frac{-3}{(2x+3y)^2} \quad (2)$$

$$w_{xy} = \frac{-6}{(2x+3y)^2}$$

$$w_{yx} = \frac{-6}{(2x+3y)^2}$$

$$w = e^u + u \ln y + y \ln u$$

$$w_x = e^u + \ln y + \frac{y}{u}, \quad w_y = \frac{u}{y} + \ln u$$

$$w_{xy} = \frac{1}{y} + \frac{1}{u}$$

$$w_{yx} = \frac{1}{u} + \frac{1}{y}$$

$$w = xy^2 + x^2y^3 + x^3y^4$$

$$w_x = y^2 + 2xy^3 + 3x^2y^4$$

$$w_y = 2xy + 3x^2y^2 + 4x^3y^3$$

$$w_{xy} = 2y + 6xy^2 + 12x^2y^3$$

$$w_{yx} = 2y + 6xy^2 + 12x^2y^3$$

$$w = u \sin y + y \sin u + y$$

$$w_x = \sin y + y \cos u + y$$

$$w_y = u \cos y + \sin u + u$$

$$w_{xy} = \cos y + \cos u + 1$$

$$w_{yx} = \cos y + \cos u + 1$$

Q 35-40

Find partial derivative w.r.t each variable.

$$f(t, \alpha) = \cos(2\pi t - \alpha)$$

$$f_t = -\sin(2\pi t - \alpha)(2\pi)$$

$$= -2\pi \sin(2\pi t - \alpha)$$

$$f_\alpha = -\sin(2\pi t - \alpha)(-1)$$

$$= \sin(2\pi t - \alpha)$$

$$g(u, v) = v^2 e^{\left(\frac{2u}{v}\right)}$$

$$g_u = v^2 e^{\left(\frac{2u}{v}\right)} \left(\frac{2}{v}\right)$$

$$= 2ve^{\frac{2u}{v}}$$

$$g_v = v^2 e^{\frac{2u}{v}} \left(-\frac{2u}{v^2}\right) + e^{\frac{2u}{v}} (2v)$$

$$= -2e^{\frac{2u}{v}} u + 2ve^{\frac{2u}{v}}$$

$$h(\rho, \phi, \theta) = \rho \sin \phi \cos \theta$$

$$h_\rho = \sin \phi \cos \theta$$

$$h_\phi = \rho \cos \phi \cos \theta$$

$$h_\theta = -\rho \sin \phi \sin \theta$$

$$g(r, \theta, z) = r(1 - \cos \theta) - z$$

$$g_r = 1 - \cos \theta$$

$$g_\theta = (r - r \cos \theta - z)'$$

$$= -r(-\sin \theta) - 0$$

$$= r \sin \theta$$

$$g_z = -1$$

Laplace Equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{OR} \quad f_{xx} + f_{yy} = 0$$

Wave Equation

$$\frac{\partial^2 w}{\partial t^2} = \frac{1}{c^2} \left(\frac{\partial^2 w}{\partial x^2} \right)$$

From Z.R. Bhatti

Exercise: 2.3

Show following functions
are satisfying Laplace
eqn.

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0.$$

57.

$$f(x, y) = \sin x \sinh y$$

$$f_x = \cos x \sinh y$$

$$f_{xx} = -\sin x \sinh y$$

$$f_y = \sin x \cosh y$$

$$f_{yy} = \sin x \sinh y$$

By Laplace eqn.

$$-\sin x \sinh y + \sin x \sinh y = 0$$

$$0 = 0$$

58.

$$f(u, y) = \cos u \cosh y$$

$$f_u = -\sin u \cosh y$$

$$f_{uu} = -\cos u \cosh y$$

$$f_y = \cos u \sinh y$$

$$f_{yy} = \cos u \cosh y$$

By Laplace eqn.

$$-\cancel{\cos u \cosh y} + \cancel{\cos u \cosh y} = 0$$

$$0 = 0$$

59.

$$f(u, y) = \ln \sqrt{u^2 + y^2}$$

$$f_u = \frac{1}{\sqrt{u^2 + y^2}} \cdot \frac{1}{2\sqrt{u^2 + y^2}} (2u)$$

$$= \frac{u}{u^2 + y^2}$$

$$f_{uu} = \frac{(u^2 + y^2)(1) - (u)(2u)}{(u^2 + y^2)^2} = \frac{u^2 + y^2 - 2u^2}{(u^2 + y^2)^2}$$

$$= \frac{-u^2 + y^2}{(u^2 + y^2)^2}$$

$$f_y = \frac{1}{\sqrt{u^2 + y^2}} \cdot \frac{1}{2\sqrt{u^2 + y^2}} (2y)$$

$$= \frac{y}{u^2 + y^2}$$

$$f_{yy} = \frac{(u^2 + y^2)(1) - (y)(2y)}{(u^2 + y^2)^2} = \frac{u^2 + y^2 - 2y^2}{(u^2 + y^2)^2}$$

$$f_{yy} = \frac{u^2 - v^2}{(u^2 + v^2)^2}$$

By Laplace eqn.

$$\frac{-u^2 + v^2}{(u^2 + v^2)^2} + \frac{u^2 - v^2}{(u^2 + v^2)^2} = 0$$

$$\frac{-\cancel{u^2} + \cancel{v^2} + u^2 - v^2}{(u^2 + v^2)^2} = 0$$

$$0 = 0.$$

60. $f(u, v) = \tan^{-1}\left(\frac{2uv}{u^2 - v^2}\right)$

$$f_y = \frac{1}{1 + \left(\frac{2uv}{u^2 - v^2}\right)^2} \cdot \left(\frac{2uv}{u^2 - v^2}\right)'$$

$$= \frac{\cancel{(u^2 - v^2)^2}}{(u^2 - v^2)^2 + (2uv)^2} \left[\frac{(\cancel{u^2 - v^2})(2u) - (2uv)(-2v)}{\cancel{(u^2 - v^2)^2}} \right]$$

$$= \frac{2u^3 - 2v^2u + 4uv^2}{(u^2 + v^2)^2}$$

$$= \frac{2u^3 + 2uv^2}{(u^2 + v^2)^2}$$

$$= \frac{2u(\cancel{u^2 + v^2})}{(u^2 + v^2)^2}$$

$$f_y = \frac{2u}{(u^2 + v^2)}$$

$$f_{yy} = \frac{(u^2+y^2)(2u)' - (2u)(u^2+y^2)'}{(u^2+y^2)^2}$$

$$= \frac{0 - (2u)(2y)}{(u^2+y^2)^2}$$

$$f_{yy} = \frac{-4uy}{(u^2+y^2)^2}$$

$$f_u = \frac{1}{1 + \left(\frac{2uy}{u^2-y^2}\right)^2} \left(\frac{2uy}{(u^2-y^2)} \right)'$$

$$= \frac{\cancel{(u^2-y^2)}}{(u^2-y^2)^2 + (2uy)^2} \left[\frac{(u^2-y^2)(2y) - (2uy)(2u)}{\cancel{(u^2-y^2)}} \right]$$

$$= \frac{(u^2-y^2)(2y) - (2uy)(2u)}{(u^2-y^2)^2 + (2uy)^2}$$

$$= \frac{2u^2y - 2y^3 - 4u^2y}{(u^2+y^2)^2}$$

$$= \frac{-2u^2y - 2y^3}{(u^2+y^2)^2}$$

$$= \frac{-2y\cancel{(u^2+y^2)}}{(u^2+y^2)^2}$$

$$= \frac{-2y}{(u^2+y^2)}$$

$$f_{xx} = \frac{(u^2+y^2)(2y)' - (-2y)(u^2+y^2)'}{(u^2+y^2)^2}$$

$$= \frac{0 - (-2y)(2u)}{(u^2+y^2)^2}$$

$$f_{xx} = \frac{4uy}{(u^2+y^2)^2}$$

By Laplace eqn.

$$-\frac{4uy}{(u^2+y^2)^2} + \frac{4uy}{(u^2+y^2)^2} = 0$$

$$0=0.$$

61.

Show $f(x,y) = \sin xy$ satisfies

$$u^2 f_{xx} - y^2 f_{yy} = 0$$

$$f(x,y) = \sin xy$$

$$f_x = y \cos xy$$

$$f_{xx} = -y^2 \sin xy$$

$$f_y = x \cos xy$$

$$f_{yy} = -x^2 \sin xy$$

acc to cdtn.

$$u^2 (-y^2 \sin xy) - y^2 (-x^2 \sin xy) = 0$$

$$-u^2 y^2 \sin xy + u^2 y^2 \sin xy = 0$$

$$0=0.$$

62. $f(x, y, z) = x^3 + 3yz + \sin yz$

Show above function satisfies $f_{xyz} = f_{zxy}$.

For f_{xyz} :

$$f_x = 3x^2 + yz \cos yz$$

$$f_{xy} = z \cos yz - yz^2 \sin yz$$

$$f_{xyz} = \cos yz - \sin(yz)(yz)(z) - 2yz \sin yz - yz^2 \cos yz (yz)$$

$$= \cos yz - yz \sin yz - 2yz \sin yz - y^2 z^2 \cos yz$$

$$f_{xyz} = \cos yz - 3yz \sin yz - y^2 z^2 \cos yz$$

For f_{zxy} :

$$f_z = 3y + yz \cos yz$$

$$f_{zx} = y \cos yz - yz^2 \sin yz$$

$$f_{zxy} = \cos yz - y \sin yz (xz) - 2yz \sin yz - yz^2 \cos yz (xz)$$

$$= \cos yz - yz \sin yz - 2yz \sin yz - y^2 z^2 \cos yz$$

$$f_{zxy} = \cos yz - 3yz \sin yz - y^2 z^2 \cos yz$$

Hence proved.

63. If $f(x, y, z, w) = \frac{xy}{z+w}$

prove $f_{xyzw} = \frac{2}{(z+w)^3}$

$$f_x = \frac{y}{z+w}$$

$$f_{xy} = \frac{1}{z+w}$$

$$f_{xyz} = \frac{-1}{(z+w)^2}$$

$$f_{xyzw} = \frac{2}{(z+w)^3}$$

Directional Derivative

Formula: $\frac{\nabla u \cdot \mathbf{v}}{|\mathbf{v}|}$

1.) Find directional derivative at (1,2)
where: $u = 2xy - \frac{y}{x}$, $\mathbf{v} = 2\hat{i} - 3\hat{j} + 0\hat{k}$.

$$\frac{\partial u}{\partial x} = 2y + \frac{y}{x^2}$$

$$\frac{\partial u}{\partial y} = 2x - \frac{1}{x}$$

$$\frac{\partial u}{\partial z} = 0$$

At Pt (1,2)

$$\nabla u_p = (6, 1, 0)$$

By Formula: $\frac{\nabla u \cdot \mathbf{v}}{|\mathbf{v}|}$

$$\Rightarrow \frac{(2\hat{i} - 3\hat{j} + 0\hat{k}) \cdot (6\hat{i} + \hat{j} + 0\hat{k})}{\sqrt{4+9}}$$

$$\Rightarrow \frac{12-3}{\sqrt{13}} \Rightarrow \frac{9}{\sqrt{13}}$$

$$2.) \quad u = y e^{-u} (u^2 + y^2 + z^2 + 1), \quad (0, 0, 0)$$

$$A = (2, 1, 2)$$

$$\frac{\partial u}{\partial x} = y \left[e^{-u} (-1) (u^2 + y^2 + z^2 + 1) + e^{-u} (2u) \right]$$

$$= y \left[-e^{-u} (u^2 + y^2 + z^2 + 1) + 2u e^{-u} \right]$$

$$\frac{\partial u}{\partial y} = e^{-u} \left[(1) (u^2 + y^2 + z^2 + 1) + y (2y) \right]$$

$$\frac{\partial u}{\partial z} = 2yz e^{-u}$$

$$\frac{\partial u}{\partial z}$$

$$\text{At } P(0, 0, 0)$$

$$(\nabla u)_P = (0, 1, 0)$$

$$ds;$$

$$\frac{(\nabla u)_P}{|\nabla u|}$$

$$|\nabla u|$$

$$= \frac{(2\hat{i} + 1\hat{j} + 2\hat{k}) (0\hat{i} + 1\hat{j} + 0\hat{k})}{\sqrt{4+1+4}}$$

$$= \frac{1}{3} ds$$

$$3$$

$$3.) U = \sinh(u+y) + \cosh z; (1, 0, 1)$$

$$V = (-2, 2, -1)$$

$$\frac{\partial U}{\partial u} = \cosh(u+y)(1) + 0$$

$$\frac{\partial U}{\partial u}$$

$$= \cosh(u+y) = \cosh(1)$$

$$= 1.5430$$

$$\frac{\partial U}{\partial y} = \cosh(u+y)(1) + 0 = \cosh(u+y)$$

$$\frac{\partial U}{\partial y}$$

$$= 1.5430$$

$$\frac{\partial U}{\partial z} = \sinh z(1) = \sinh(1)$$

$$\frac{\partial U}{\partial z}$$

$$= 1.1752$$

$$\text{At Pt } (1, 0, 1)$$

$$(\nabla)P = (1.54, 1.54, 1.17)$$

Ans;

$$\frac{V(\nabla U)_P}{|V|}$$

$$|V|$$

$$(-2\hat{i} + 2\hat{j} - 1\hat{k})(1.54\hat{i} + 1.54\hat{j} + 1.17\hat{k})$$

$$\sqrt{4+4+1}$$

$$= \frac{-3.086 + 3.086 - 1.1752}{3}$$

$$3$$

$$= \frac{-1.1752}{3}$$

$$3$$

$$4.) \quad g = \cos uy + e^{yz} + \ln zu, \quad (1, 0, \frac{1}{2})$$

$$A = (\hat{i} + 2\hat{j} + 2\hat{k})$$

$$\frac{\partial g}{\partial u} = -\sin uy \cdot y + 0 + \frac{1}{z}$$

$$= -y \sin uy + 1 = 1$$

$$\frac{\partial g}{\partial y} = -\sin uy (u) + e^{yz} \cdot z + 0$$

$$= -u \sin uy + e^{yz} \cdot z = 0 + 1 = 1$$

$$\frac{\partial g}{\partial z} = e^{yz} \cdot y + \frac{1}{z} \cdot \frac{1}{z} = ye^{yz} + \frac{1}{z}$$

$$= \frac{1}{1(2)} = \frac{1}{2}$$

$$\text{At Pt } (1, 0, \frac{1}{2})$$

$$(\nabla U)_P = (1, \frac{1}{2}, 2)$$

$$= \frac{(1\hat{i} + 2\hat{j} + 2\hat{k}) (1\hat{i} + \frac{1}{2}\hat{j} + 2\hat{k})}{\sqrt{1+4+4}}$$

$$= \frac{1+1+4}{\sqrt{9}}$$

$$= \frac{2\cancel{6}}{\cancel{3}} = 2$$

Exercise 15.1

Finding regions of integration
and double integrals.

Q(1-10)

1.) $\int_0^3 \int_0^2 (4 - y^2) dy dx$

$$\int_0^3 \left[4y - \frac{y^3}{3} \right]_0^2 dx$$

$$\int_0^3 \left\{ 4(2) - 4(0) - \frac{1}{3} ((2)^3 - (0)^3) \right\} dx$$

$$\int_0^3 \left(8 - \frac{8}{3} \right) dx$$

$$\int_0^3 \frac{16}{3} dx$$

$$\left[\frac{16}{3} x \right]_0^3 \Rightarrow \frac{16}{3} (3-0) = \frac{16}{3} (3)$$

$$\Rightarrow 16$$

2.) $\int_0^3 \int_{-2}^0 (x^2 y - 2xy) dy dx$

$$\int_0^3 \left[\frac{x^2 y^2}{2} - \frac{2xy^2}{2} \right]_{-2}^0 dx$$

$$\int_0^3 \left\{ \frac{1}{2} x^2 (0 - (-4)) - x(0 - 4) \right\} dx$$

$$\int_0^3 \left(\frac{-4u^2 + 4u}{x} \right) du$$

$$\int_0^3 4u - 2u^2 du$$

$$\Rightarrow \left| \frac{4u^2}{2} - \frac{2u^3}{3} \right|_0^3$$

$$\Rightarrow 2(9-0) - \frac{2}{3}(27-0)$$

$$\Rightarrow 18-18$$

$$\Rightarrow 0$$

3.)

$$\int_{-1}^0 \int_{-1}^1 (u+y+1) du dy$$

$$\int_{-1}^0 \left\{ \left| \frac{u^2}{2} + uy + u \right|_{-1}^1 \right\} dy$$

$$\int_{-1}^0 \left\{ \frac{1}{2} (1-1) + y(1+1) + (1+1) \right\} dy$$

$$\Rightarrow \int_{-1}^0 (0+2y+2) dy$$

$$\Rightarrow \left| \frac{2y^2}{2} + 2y \right|_{-1}^0$$

$$\Rightarrow 0-1+2(0+1)$$

$$-1+2 \Rightarrow 1$$

4.)

$$\int_{\pi}^{2\pi} \int_0^{\pi} (\sin u + \cos y) du dy$$

$$\Rightarrow \int_{\pi}^{2\pi} \left\{ -\cos u + u \cos y \right\}_0^{\pi} dy$$

$$\Rightarrow \int_{\pi}^{2\pi} -\cos(\pi) + \cos(0) + \pi \cos y - 0 dy$$

$$\int_{\pi}^{2\pi} 1 + 1 + \pi \cos y dy$$

$$\Rightarrow \int_{\pi}^{2\pi} 2 + \pi \cos y dy$$

$$\Rightarrow \left| 2y + \pi \sin y \right|_{\pi}^{2\pi}$$

$$= 2(2\pi - \pi) + (\pi \sin(2\pi) - \pi \sin(\pi))$$

$$\Rightarrow 2\pi + 0$$

$$= 2\pi \text{ u.}$$

5.)

$$\int_0^{\pi} \int_0^u u \sin y dy du$$

$$\Rightarrow \int_0^{\pi} -u \cos y du$$

$$\Rightarrow \int_0^{\pi} -u (\cos(u) - \cos(0)) du$$

$$\Rightarrow \int_0^{\pi} (-u \cos u + u) du$$

$$\Rightarrow \left| \left[-u \sin u + \int (-1) \sin u du \right] + \frac{u^2}{2} \right|_0^{\pi}$$

$$\Rightarrow \left| -u \sin u - \cos u + \frac{u^2}{2} \right|_0^{\pi}$$

$$\Rightarrow \left| -u \sin u - \cos u + \frac{u^2}{2} \right|_0^{\pi}$$

$$\Rightarrow \left[-\pi \sin \pi - \cos \pi + \frac{\pi^2}{2} - (0 + 1 + 0) \right]$$

$$\Rightarrow 0 + 2 + \frac{\pi^2}{2}$$

$$= \frac{4 + \pi^2}{2} \text{ Ans.}$$

6.)

$$\int_0^{\pi} \int_0^{\sin u} y dy du$$

$$\Rightarrow \int_0^{\pi} \left| \frac{y^2}{2} \right|_0^{\sin u} du$$

$$\Rightarrow \int_0^{\pi} \frac{1}{2} (\sin^2 u - 0) du$$

$$\Rightarrow \int_0^{\pi} \frac{1}{2} \left(\frac{1 - \cos 2u}{2} \right) du$$

$$\Rightarrow \frac{1}{4} \int_0^{\pi} 1 - \cos 2u du$$

$$\Rightarrow \frac{1}{4} \left| u - \frac{\sin 2u}{2} \right|_0^{\pi}$$

$$\Rightarrow \frac{1}{4} \left[\left[\frac{\pi - \sin 2\pi}{2} \right] - \left[\frac{0 - \sin 2(0)}{2} \right] \right]$$

$$\Rightarrow \frac{1}{4} \left[\frac{\pi - \sin 2\pi}{2} \right]$$

$$\Rightarrow \frac{1}{4} (\pi)$$

$$\Rightarrow \frac{\pi}{4}$$

7.) $\int_1^{\ln 8} \int_0^{\ln y} e^{xy} dy dx$

$$\Rightarrow \int_1^{\ln 8} \left[e^{xy} \right]_0^{\ln y} dy$$

$$\Rightarrow \int_1^{\ln 8} [\ln y]$$

$$\Rightarrow \int_1^{\ln 8} (e^{\ln y + 1} - e^{0+1}) dy$$

$$\Rightarrow \int_1^{\ln 8} [ye^y - e^y] dy$$

$$\Rightarrow \int_1^{\ln 8} \left\{ \left[ye^y - \int_1^y e^y dy \right] - e^y \right\} dy$$

$$= \left[ye^y - e^y - e^y \right]_1^{\ln 8}$$

$$= \left[(e^y (y-1) - e^y) \right]_1^{\ln 8}$$

$$\Rightarrow \left[e^{\ln 8} (\ln 8 - 1) - e^{\ln 8} - (e'(1-1) - e') \right]$$

$$= 8(\ln 8 - 1) - 8 + e$$

$$= 8 \ln 8 - 16 + e.$$

$$8.) \int_1^2 \int_y^{y^2} du \, dy$$

$$\Rightarrow \int_1^2 \ln|y|^{y^2} dy$$

$$\Rightarrow \int_1^2 (y^2 - y) dy$$

$$\Rightarrow \left| \frac{y^3}{3} - \frac{y^2}{2} \right|_1^2$$

$$= \frac{1}{3} (8-1) - \frac{1}{2} (4-1)$$

$$= \frac{7}{3} - \frac{3}{2} \Rightarrow \frac{5}{6}$$

$$9) \int_0^1 \int_0^y 3y^2 e^{xy} dx dy$$

$$\Rightarrow \int_0^1 \left[3y^2 \frac{e^{xy}}{y} \right]_0^y dy$$

$$\Rightarrow \int_0^1 3y e^{xy} \Big|_0^y dy$$

$$\Rightarrow \int_0^1 (3y^2 e^{y^2} - 3y^2 e^{0 \cdot y}) dy$$

$$= \int_0^1 (3y^2 e^{y^2} - 3y^2) dy$$

$$\Rightarrow \left[\frac{3y^3}{3} \cdot \frac{e^{y^2}}{2y} - \int \frac{3y^2}{3y^2} \right]_0^1 = \left[\frac{y^3}{2} e^{y^2} - y \right]_0^1$$

$$= \left[\frac{1}{2} e^1 - 1 \right]_0^1$$

$$= \left[\left(\frac{1}{2} e - 1 \right) - \left(\frac{1}{2} e^0 - 1 \right) \right]$$

$$= \frac{e}{2} - 1 - \frac{1}{2} + 1$$

$$= \frac{e-2}{2}$$

$$10.) \int_1^4 \int_0^{\sqrt{u}} \frac{3}{2} e^{y/\sqrt{u}} dy du$$

$$\Rightarrow \frac{3}{2} \int_1^4 \left| \sqrt{u} e^{\frac{y}{\sqrt{u}}} \right|_0^{\sqrt{u}} du$$

$$= \frac{3}{2} \int_1^4 \left| \sqrt{u} e^{\frac{\sqrt{u}}{\sqrt{u}}} - \sqrt{u} e^{\frac{0}{\sqrt{u}}} \right| du$$

$$= \frac{3}{2} \int_1^4 \sqrt{u} (e - 1) du$$

$$= \frac{3}{2} (e-1) \int_1^4 \sqrt{u} du$$

$$= \frac{3(e-1)}{2} \left| \frac{2}{3} u^{3/2} \right|_1^4$$

$$= \frac{3(e-1)}{2} \frac{2}{3} \left| u^{3/2} \right|_1^4$$

$$= (e-1) (4^{3/2} - 1)$$

$$= (e-1) (2^{2 \cdot 3/2} - 1)$$

$$= 7(e-1) \text{ d.}$$

117-38)

17)

$$\int_{-2}^0 \int_v^{-v} 2 \, dp \, dv$$

$$\Rightarrow \int_{-2}^0 \left[2p \right]_v^{-v} dv$$

$$\Rightarrow \int_{-2}^0 2(-v-v) dv$$

$$\Rightarrow 2 \int_{-2}^0 -2v dv$$

$$\Rightarrow -4 \left[\frac{v^2}{2} \right]_{-2}^0 \Rightarrow -2 \left[(0)^2 - (-2)^2 \right]$$

$$\Rightarrow -2(-4) = +8$$

18)

$$\int_0^1 \int_0^{\sqrt{1-s^2}} 8t \, dt \, ds$$

$$\Rightarrow \int_0^1 \left[\frac{8t^2}{2} \right]_0^{\sqrt{1-s^2}} ds$$

$$\Rightarrow 4 \int_0^1 \left[(\sqrt{1-s^2})^2 - (0)^2 \right] ds$$

$$\Rightarrow 4 \int_0^1 (1-s^2) ds \Rightarrow 4 \left[s - \frac{s^3}{3} \right]_0^1$$

$$\Rightarrow 4 \left[(1-0) - \frac{1}{3}(1-0) \right] = 4 \left(\frac{2}{3} \right) = \frac{8}{3}$$

19)

$$\begin{aligned}
 & \int_{-\pi/3}^{\pi/3} \int_0^{\sec t} 3 \cos t \, du \, dt \\
 \Rightarrow & 3 \int_{-\pi/3}^{\pi/3} \left| u \cos t \right|_0^{\sec t} dt \\
 \Rightarrow & 3 \int_{-\pi/3}^{\pi/3} \sec t \cos t \, dt \\
 \Rightarrow & 3 \int_{-\pi/3}^{\pi/3} \frac{1}{\cancel{\cos t}} \cos t \, dt \\
 \Rightarrow & 3 \left| t \right|_{-\pi/3}^{\pi/3} \\
 \Rightarrow & 3 \left[\frac{\pi}{3} + \frac{\pi}{3} \right] \Rightarrow \frac{2\pi}{1}
 \end{aligned}$$

$$= 2\pi \text{ Ans.}$$

20)

$$\begin{aligned}
 & \int_0^3 \int_1^{4-2v} \frac{4-2u}{v^2} \, du \, dv \\
 \Rightarrow & \int_0^3 \int_1^{4-2v} \left(\frac{4}{v^2} - \frac{2u}{v^2} \right) \, du \, dv \\
 \Rightarrow & \int_0^3 \left| -\frac{4}{v} + \frac{2u}{v} \right|_1^{4-2v} dv \\
 \Rightarrow & \int_0^3 \left[\frac{-4}{4-2v} + \frac{2v}{4-2v} - \left(\frac{-4}{1} + \frac{2v}{1} \right) \right] dv \\
 \Rightarrow & \int_0^3 \left[\frac{2v-4+(4-2v)}{4-2v} \right] dv
 \end{aligned}$$

$$\Rightarrow \int_0^3 -1 + 4 - 2u \, du$$

$$\Rightarrow \left| 3u - \frac{2u^2}{2} \right|_0^3$$

$$= 3(3-0) - (9-0)$$

$$= 9 - 9$$

$$= 0 \text{ hrs.}$$

$$21) \int_0^1 \int_2^{4-2x} dy \, dx$$

$$= \int_0^1 \left| y \right|_2^{4-2x} dx$$

$$= \int_0^1 [4-2x-2] dx \Rightarrow \int_0^1 (2-2x) dx$$

$$= 2 \int_0^1 (1-x) dx \Rightarrow 2 \left[\left| x - \frac{x^2}{2} \right|_0^1 \right]$$

$$= 2 \left[(1-0) - \frac{1}{2} (1-0) \right]$$

$$= 2 \left[1 - \frac{1}{2} \right] \Rightarrow 2 - \frac{1}{2} \Rightarrow 1$$

$$22 \int_0^2 \int_{y-2}^0 dx \, dy$$

$$= \int_0^2 \left| x \right|_{y-2}^0 dy$$

$$\Rightarrow \int_0^2 0 - (y-2) dy$$

$$\Rightarrow \int_0^2 (2-y) dy$$

$$= \left| 2y - \frac{y^2}{2} \right|_0^2 \Rightarrow \frac{2(2-0) - \frac{1}{2}(4-0)}{2}$$

$$= 4 - 2$$

$$= 2$$

$$23) \int_0^1 \int_y^{\sqrt{y}} dx dy$$

$$\Rightarrow \int_0^1 \left| x \right|_y^{\sqrt{y}} dy \Rightarrow \int_0^1 (\sqrt{y} - y) dy$$

$$\Rightarrow \left| \frac{2}{3} y^{3/2} - \frac{y^2}{2} \right|_0^1 \Rightarrow \frac{2}{3} ((1)^{3/2} - 0) - \frac{1}{2} ((1)^2 - 0)$$

$$= \frac{2}{3} (1) - \frac{1}{2} \Rightarrow \frac{2}{3} - \frac{1}{2} \Rightarrow \frac{4-3}{6} = \frac{1}{6}$$

$$24) \int_0^1 \int_{1-x}^{1-x^2} dy dx$$

$$\Rightarrow \int_0^1 \left| y \right|_{1-x}^{1-x^2} dx \Rightarrow \int_0^1 (1-x^2) - (1-x) dx$$

$$\Rightarrow \int_0^1 (x - x^2 + x) dx \Rightarrow \int_0^1 (2x - x^2) dx$$

$$\Rightarrow \left| \frac{x^2}{2} - \frac{x^3}{3} \right|_0^1 \Rightarrow \frac{1}{2} (1-0) - \frac{1}{3} (1-0)$$

$$= \frac{1}{2} - \frac{1}{3} \Rightarrow \frac{3-2}{6} \Rightarrow \frac{1}{6}$$

$$25) \int_0^1 \int_1^{e^x} dy dx$$

$$\Rightarrow \int_0^1 |y|_1^{e^x} dx$$

$$\Rightarrow \int_0^1 (e^x - 1) dx \Rightarrow |e^x - x|$$

$$\Rightarrow (e^1 - 1) - (1 - 0)$$

$$= e - 1 - 1$$

$$= e - 2$$

$$26) \int_0^{\ln 2} \int_0^{e^y} dx dy$$

$$= \int_0^{\ln 2} |x|_0^{e^y} dy$$

$$\Rightarrow \int_0^{\ln 2} (e^y - 0) dy \Rightarrow |2y - ye^y|_0^{\ln 2}$$

$$\Rightarrow 2(\ln 2 - 0) - e^{\ln 2}(\ln 2 - 0)$$

$$= 2\ln 2 - e^{\ln 2} \ln 2$$

$$27) \int_0^{3/2} \int_0^{9-4x^2} 16xy dx dy$$

$$\Rightarrow \int_0^{3/2} |16xy|_0^{9-4x^2} dy$$

$$\Rightarrow \int_0^{3/2} 16u(9-4u^2-0)du$$

$$= 16 \int_0^{3/2} (9u-4u^3)du$$

$$\Rightarrow 16 \left| \frac{9u^2}{2} - \frac{4u^4}{4} \right|_0^{3/2}$$

$$= 16 \left[\frac{9}{2} \left(\left(\frac{3}{2} \right)^2 - 0 \right) - \left(\left(\frac{3}{2} \right)^4 - 0 \right) \right]$$

$$= 16 \left[\frac{9}{2} \cdot \frac{9}{4} - \frac{81}{16} \right]$$

$$\Rightarrow 16 \left[\frac{81}{8} - \frac{81}{16} \right]$$

$$\Rightarrow 2(81) - 81 \Rightarrow 81$$

28) $\int_0^2 \int_0^{4-y^2} y dx dy$

$$\Rightarrow \int_0^2 \left[xy \right]_0^{4-y^2} dy \Rightarrow \int_0^2 y(4-y^2-0) dy$$

$$\Rightarrow \int_0^2 (4y - y^3) dy \Rightarrow \left| \frac{4y^2}{2} - \frac{y^4}{4} \right|_0^2$$

$$\Rightarrow \frac{2(4-0) - \frac{1}{4}((2)^4-0)}{4} = \frac{8-16}{4}$$

$$= 4$$

$$29.) \int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} 3y \, dx \, dy$$

$$\Rightarrow \int_0^1 \left[3xy \right]_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} dy$$

$$\Rightarrow \int_0^1 3y(\sqrt{1-y^2} + \sqrt{1-y^2}) dy$$

$$\Rightarrow 3 \int_0^1 (y\sqrt{1-y^2} + y\sqrt{1-y^2}) dy$$

$$\Rightarrow 3(2) \int_0^1 y\sqrt{1-y^2} dy$$

$$\Rightarrow 6 \int_0^1 \frac{y}{(1-y^2)^{1/2}} dy \Rightarrow \frac{36}{-2} \int \frac{-2y}{(1-y^2)^{1/2}}$$

$$= -3 \left| \frac{(1-y^2)^{1/2+1}}{3/2} \right|_0^1 \Rightarrow \frac{2}{3} \cdot (-3) [(1-1) - (1-0)]$$

$$= -2(-1) = 2$$

$$30.) \int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 6xy \, dy \, dx$$

$$\Rightarrow \int_0^2 \left[6xy \right]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx$$

$$\Rightarrow \int_0^2 6x(\sqrt{4-x^2} + \sqrt{4-x^2}) dx$$

$$\Rightarrow \int_0^2 6x(2\sqrt{4-x^2}) dx \Rightarrow \int_0^2 12x\sqrt{4-x^2} dx$$

$$= -6 \int_0^2 \sqrt{4-u^2} (-2u) du \Rightarrow \frac{-6}{3} \left| (4-u^2)^{3/2} \right|_0^2$$

$$= -4 \left[(4-2)^{3/2} - (4-0)^{3/2} \right]$$

$$\Rightarrow -4 \left[(2)^{3/2} - (2)^{2 \cdot 3/2} \right]$$

$$\Rightarrow -4 \left[2 \cdot 828 - 8 \right] \Rightarrow 20.8$$

m. Tahakhan.